Towards an Intelligent Video Filtering System for Age-Appropriate Media Consumption*

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Abstract

The rapid growth of online video platforms has increased the need for effective mechanisms to protect young audiences from exposure to inappropriate content. Traditional manual moderation approaches are often labor-intensive, costly, and difficult to scale to the large volumes of multimedia content uploaded daily. This study proposes an AI-based automated video screening system for youth-appropriate media consumption using a hybrid deep learning framework that combines a pretrained ResNet-50 Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) network. The proposed architecture leverages ResNet-50 to extract high-level spatial features from video frames, while the LSTM captures temporal dependencies across sequential frames to improve contextual understanding of video content. The system is integrated into a browser extension that performs near real-time content moderation through server-side inference and local caching of classification results using IndexedDB. Experimental evaluation was conducted on a combined dataset consisting of YouTube-8M, NSFW video samples, and a custom youth-oriented dataset containing safe and unsafe content. Results demonstrate that the proposed ResNet50–LSTM framework outperforms conventional keyword-based and metadata-based filtering approaches, achieving 93.2% precision, 91.6% recall, 92.4% F1-score, and 92.3% accuracy. The findings highlight the effectiveness of combining spatial and temporal visual analysis for automated content moderation. This work contributes to the development of scalable,

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reliable, and practically deployable AI-driven moderation systems that support safer digital media environments for youth audiences.

1 Introduction

The rapid growth of online video platforms has fundamentally transformed how young audiences consume digital media. Children and adolescents increasingly rely on streaming services, video-sharing platforms, and social media for entertainment, education, and social interaction. However, the vast availability of user-generated and professionally produced multimedia content presents a significant challenge: protecting youth from exposure to inappropriate, harmful, or age-restricted material. Traditional content moderation approaches, including manual review and keyword-based filtering, are often insufficient due to the enormous scale, diversity, and continuously evolving nature of online video content.

Automated video screening systems have emerged as a promising solution to this challenge. Recent advances in artificial intelligence (AI), particularly in computer vision, natural language processing (NLP), and multimodal learning, enable automated systems to analyze multiple dimensions of video content, including visual frames, spoken language, textual metadata, and contextual information. Unlike conventional rule-based methods, AI-driven approaches can identify more nuanced forms of inappropriate content, such as violent imagery, offensive language, explicit scenes, harmful behaviors, and implicit contextual risks. Moreover, such systems can operate at scale and near real time, making them highly suitable for modern digital media platforms where vast volumes of content are uploaded continuously.

From the perspective of E-Business Information Systems, intelligent content moderation plays an increasingly critical role in supporting sustainable and trustworthy digital ecosystems. Online video platforms are not only media distribution channels but also major digital business infrastructures that rely on user trust, regulatory compliance, and responsible platform governance. AI-powered moderation systems contribute to platform safety, enhance user experience, reduce operational moderation costs, and support age-appropriate recommendation mechanisms. As digital businesses increasingly adopt AI-driven decision-making systems, ensuring responsible governance, fairness, transparency, and accountability in automated moderation becomes essential.

This study proposes an AI-based automated video filtering system for youth-appropriate media consumption using a hybrid deep learning framework for visual content moderation. The proposed architecture integrates a pretrained Convolutional Neural Network (CNN) based on ResNet-50 for spatial feature extraction and a Recurrent Neural Network (RNN) using Long Short-Term Memory (LSTM) for temporal sequence modeling of video frames. This hybrid architecture enables the system to capture both frame-level visual semantics and temporal contextual dependencies, improving the detection of inappropriate content such as violence, explicit imagery, harmful behaviors, and disturbing scenes. Compared with traditional keyword-based and metadata-only filtering approaches, the proposed framework provides significantly enhanced detection capability for visually complex and context-sensitive inappropriate content scenarios.

Beyond technical performance, this research also recognizes important practical and ethical considerations associated with AI-driven moderation systems. False positives may result in the unintended blocking of legitimate educational, entertainment, or culturally sensitive content, while

biases in training datasets may lead to unfair classification outcomes across different content domains, including animated or culturally specific media. Furthermore, explainability remains a critical concern, as platform operators, regulators, parents, and users increasingly require transparent and accountable decision-making from automated moderation systems that influence content accessibility. Addressing these concerns is essential to maintaining an appropriate balance between youth protection, freedom of expression, inclusivity, and fairness within digital media ecosystems.

Experimental evaluation using benchmark multimedia datasets and a custom youth-oriented video dataset demonstrates that the proposed ResNet50–LSTM hybrid framework achieves superior classification performance compared with conventional baseline approaches, including keyword-based filtering, metadata-based screening, and single-model architectures. The results show substantial improvements in precision, recall, F1-score, and overall accuracy, highlighting the effectiveness of combining spatial and temporal visual analysis for automated content moderation.

This work contributes to the advancement of intelligent E-Business Information Systems by presenting a practically deployable AI moderation framework capable of supporting safer digital media consumption, responsible platform governance, and scalable automated content screening. The integration of browser-based deployment with server-side AI inference and local caching mechanisms further demonstrates the feasibility of implementing efficient, near real-time moderation solutions for modern online video ecosystems.

2 Related Theory

2.1 Theoretical Foundations of AI-Based Video Content Moderation

The development of an Automated Video Screening System for Youth-Appropriate Content is grounded in several interdisciplinary theoretical and technological domains, including media regulation, computer vision, machine learning, and automated content moderation. These foundations provide the conceptual and technical basis for detecting and classifying inappropriate video content that may negatively affect young audiences.

Content moderation on digital platforms is strongly influenced by media regulation frameworks and child protection guidelines, which emphasize safeguarding children from exposure to violent, explicit, or otherwise harmful material [5]. Traditional moderation approaches, such as manual review and keyword-based filtering, often struggle to scale effectively and frequently fail to identify contextual or visually implicit harmful content. Consequently, automated moderation systems have emerged as a practical solution for large-scale content screening, supporting age-appropriate content classification and safer digital environments.

A core enabling technology of automated video moderation is computer vision, which allows machines to interpret and analyze visual information contained in video frames [6], [7]. Modern computer vision systems are primarily based on deep learning architectures, particularly Convolutional Neural Networks (CNNs), which can automatically learn hierarchical representations of objects, actions, textures, and scene characteristics. Well-established CNN architectures, including VGGNet, ResNet, and Inception, have demonstrated strong performance in image and video classification tasks. In the proposed system, a pretrained ResNet-50 model serves as the primary visual feature extractor, enabling the identification of potentially inappropriate visual content such as violence, nudity, harmful behaviors, and disturbing imagery.

Since video content consists of sequential frames rather than isolated images, temporal information must also be considered. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory

(LSTM) networks, are widely used for modeling temporal dependencies and motion patterns across video sequences. By analyzing frame-to-frame relationships, LSTM networks can capture contextual behaviors that may not be observable from individual frames alone, thereby improving the detection of unsafe visual events.

The classification process is further grounded in supervised machine learning theory, where models learn discriminative patterns from labeled training data to predict whether content is safe or unsafe [3], [11]. Numerous machine learning algorithms, including Support Vector Machines (SVMs), Random Forests, and deep neural networks, have been successfully applied to multimedia classification and content moderation tasks. The proposed hybrid ResNet50–LSTM architecture combines spatial feature learning with temporal sequence modeling to improve classification accuracy and robustness. Furthermore, the precision–recall tradeoff is an important consideration in content moderation systems, as maximizing harmful content detection must be balanced against minimizing false positives that could unnecessarily restrict legitimate content.

Although video content may contain visual, audio, and textual information, the current study focuses primarily on visual content analysis. Nevertheless, the broader field of multimedia moderation is influenced by multimodal learning principles, which suggest that combining multiple information sources can provide richer contextual understanding and improve classification performance [9]–[11]. Future enhancements may incorporate audio signals, speech transcripts, and metadata to further improve moderation accuracy and contextual awareness.

Collectively, these theoretical foundations support the development of an intelligent video moderation framework capable of automatically identifying inappropriate content while maintaining scalability, reliability, and practical applicability for youth-oriented digital media platforms.

2.2 Related work

The Internet has become an essential part of children's daily lives, serving as a major source of entertainment, education, and social engagement through online video and streaming platforms [2]. However, despite the growing availability of child-oriented digital services, many platforms continue to host inappropriate or harmful content, often with insufficient regulatory oversight. Traditional manual content review is not only labor-intensive and costly but also impractical given the massive volume of continuously uploaded multimedia content [1]. To address this challenge, previous research introduced The Eye, a cloud-based streaming platform designed to enhance parental control by allowing customized content flagging for safer, age-appropriate viewing experiences [2]. The system employed a hybrid Convolutional Neural Network–Recurrent Neural Network (CNN–RNN) architecture trained on a specialized dataset for inappropriate content classification, achieving an accuracy of 67% on test data while demonstrating scalability through deployment on Amazon Web Services (AWS) cloud infrastructure [2].

As children's dependence on online platforms continues to grow, specialized services such as YouTube Kids have emerged to provide safer digital environments for younger audiences [4]. Although such platforms operate under child protection guidelines and content regulations, inappropriate material can still bypass filtering mechanisms. Conventional moderation approaches on these platforms often rely heavily on metadata attributes such as titles, tags, or descriptions, which can be easily manipulated or insufficient for accurately identifying harmful content. To overcome these limitations, more advanced real-time video screening methods have been proposed. One promising approach applies the thin-slicing theory, in which short video and audio segments are randomly sampled for efficient yet representative analysis. This method enables real-time moderation while preserving temporal coverage across video content. Audio segments are processed through Automatic Speech Recognition (ASR) to transcribe

spoken language for linguistic analysis, while acoustic event detection identifies contextual cues such as explosions, aggressive speech, or other potentially harmful sounds. Simultaneously, sampled video frames undergo visual analysis to detect violent scenes, explicit imagery, or other inappropriate content. This multimodal analysis framework represents a significant advancement over traditional metadata-based filtering by integrating visual, audio, and textual intelligence to improve detection robustness and contextual understanding [4].

Building upon these advancements, this research aims to design and implement an automated video screening system for youth-appropriate content using advanced multimedia analysis, contextual content understanding, and AI-driven decision-making. The proposed system focuses on leveraging multimodal learning techniques to improve content moderation accuracy while incorporating ethical considerations such as fairness, contextual sensitivity, and responsible AI governance. Ultimately, this study seeks to contribute to the development of scalable, adaptive, and reliable digital safety solutions that support safer media consumption for youth audiences across diverse online content ecosystems.

3 Methodology

3.1 Definition of Inappropriate Content and System Overview

In this study, inappropriate or unsafe video content refers to any visual, auditory, or textual material that may negatively affect young audiences, based on international media classification principles and Thailand's Film and Video Act. For the purposes of dataset annotation and system classification, such content is labeled as unsafe. The labeling criteria include several categories of harmful content: violence, characterized by the presence of blood, physical aggression, weapons, explosions, or depictions of harm; nudity and sexual content, including partial or full nudity, sexual gestures, or adult-themed material; offensive language, such as profanity, hate speech, or verbally abusive expressions identified through transcript analysis; substance use and harmful behaviors, including smoking, alcohol consumption, or self-harm actions; and fear-inducing or disturbing imagery, such as horror scenes, threatening situations, or distressing visual effects deemed unsuitable for children. To ensure labeling consistency and reliability, video segments were independently annotated by multiple reviewers, achieving an inter-annotator agreement exceeding 85% Cohen's κ across visual and audio modalities.

Based on this content definition, the proposed automated video screening system is designed to detect and block inappropriate content on online video platforms, with YouTube serving as the primary use case. The overall system architecture consists of four major components: the target video platform, a browser extension, a local database, and a classification server. The target platform, represented by YouTube, serves as the source of video content for analysis. The browser extension functions as the user-facing component responsible for monitoring accessed video content, retrieving metadata such as the video identifier and title, and coordinating communication with the classification system. It also modifies the platform interface to restrict access when unsafe content is detected.

To improve efficiency and reduce redundant computation, the system utilizes Chrome IndexedDB as a local storage mechanism for caching previously analyzed video identifiers and their corresponding classification labels. When a user accesses a video, the extension first checks the local database to determine whether the content has already been processed. If a prior classification result exists, the stored decision is applied immediately without requiring server-side analysis.

If the video has not been previously analyzed, relevant content information is transmitted to the server, which performs the core classification process using an AI-based content moderation model. The

server applies the trained deep learning classifier to assess whether the content is appropriate for youth audiences based on the predefined unsafe content criteria. Once classification is complete, the resulting label is returned to the browser extension and stored in the local database for future reference. If the content is classified as unsafe, the extension automatically blocks playback by altering the webpage interface. The overall architecture enables efficient, scalable, and near real-time automated content moderation for safer digital media consumption. Details of the proposed framework are illustrated in Figure 1.

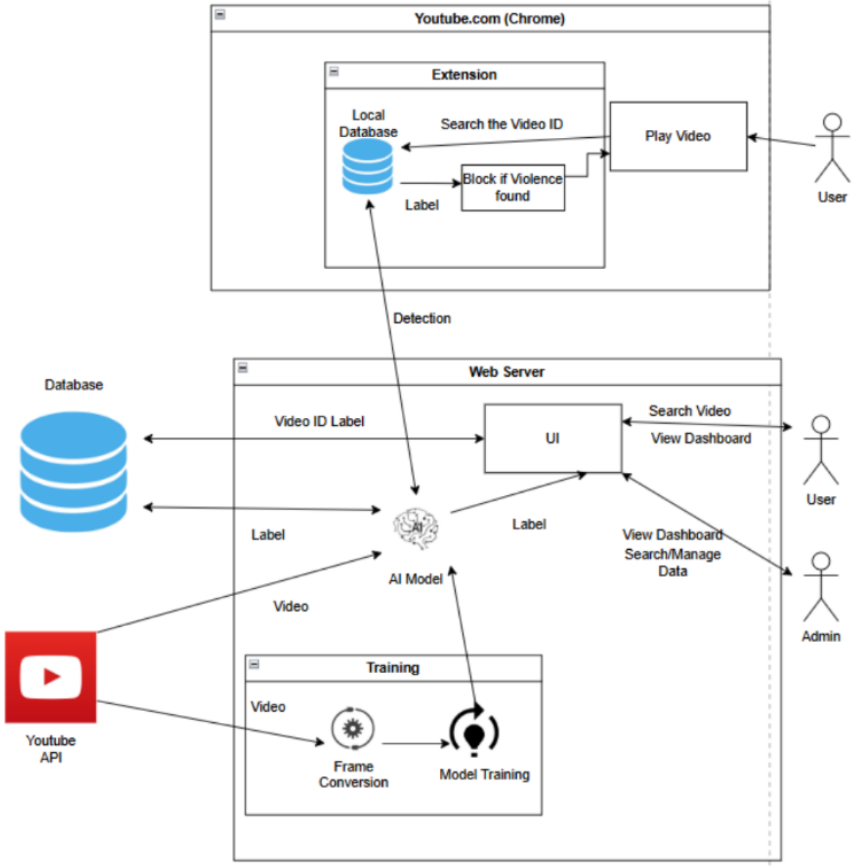


Figure 1: The proposed system

3.2 Framework for model training and evaluation

This study employs a hybrid deep learning framework that combines a pretrained Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN) for automated video content classification. Specifically, the proposed model utilizes ResNet-50 as the visual feature extraction backbone to capture high-level spatial representations from video frames, while a Long Short-Term Memory (LSTM) network is used to model temporal dependencies across sequential frames within each video segment. The primary objective is to detect inappropriate visual content for youth audiences, including violent, explicit, or harmful scenes, through efficient analysis of short video clips. To support practical deployment in browser-based content moderation, the system is designed with computational

optimization mechanisms, including server-side inference and local caching of classification results, enabling near real-time automated screening while minimizing redundant processing overhead.

The overall framework consists of five major stages: (1) data collection and labeling, (2) preprocessing, (3) feature extraction and temporal modeling, (4) model training, and (5) evaluation and system integration.

1. **Data Collection and Labeling**. Video clips were collected from publicly available multimedia datasets, including YouTube-8M, NSFW datasets, and a custom youth-oriented dataset. Each video segment was manually annotated as either safe or unsafe according to predefined content moderation criteria based on youth protection guidelines, including violence, nudity, offensive behavior, and harmful visual content.
2. **Preprocessing**. To standardize input data, all videos were resized to a spatial resolution of 224 × 224 pixels and resampled at 25 frames per second (fps). Each video was divided into non-overlapping 1-second clips, producing sequences of 25 consecutive frames per sample. To improve generalization and robustness, data augmentation techniques were applied during training, including random horizontal flipping, brightness adjustment, and rotation transformation. These preprocessing steps help reduce overfitting and improve model performance across diverse video scenarios.
3. **Feature extraction.** The proposed classification framework combines spatial feature extraction with temporal sequence learning. **Spatial feature extraction**, for visual analysis, each video frame is processed using a pretrained ResNet-50 convolutional neural network, initialized with ImageNet weights.ResNet-50 was selected due to its strong performance in visual recognition tasks and its residual learning architecture, which enables effective extraction of high-level semantic features while mitigating vanishing gradient issues.For each input frame: size: 224 × 224 × 3, output feature vector: 2048-dimensional extracted from the global average pooling layer. These frame-level feature vectors capture important spatial cues associated with inappropriate visual content, including violent scenes, explicit imagery, and harmful activities. **Temporal sequence modeling**, because video content contains temporal dependencies across frames, sequential modeling is performed using a Long Short-Term Memory (LSTM) network. The LSTM receives the ordered sequence of frame-level feature vectors and learns temporal contextual patterns that may indicate unsafe behavior over time. LSTM configuration composed of these properties, hidden units: 128, dropout rate: 0.3, sequence length: 25 frames. The final hidden representation is passed to a dense classification layer with softmax activation to predict which are Safe and Unsafe.
4. **Model training**. The hybrid ResNet50–LSTM framework was trained end-to-end using the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and 50 epochs. The categorical cross-entropy loss function was used to minimize classification error. Early stopping was applied based on validation loss to avoid overfitting. All models were implemented using TensorFlow 2.15. The hybrid ResNet50–LSTM is trained on 80% of the data, validated on 10%, and tested on the remaining 10%. Cross-validation ensures robustness.
5. **Evaluation & integration** — Metrics (Precision, Recall, F1, Accuracy) are computed. The trained model is integrated into the extension for real-time screening. After training, the optimized model was deployed as the backend inference engine for the browser extension. When a user accesses video content: (1) the extension checks IndexedDB for prior classification results. (2) if unavailable, video metadata is sent to the backend.(3) representative video frames are analyzed by the ResNet50–LSTM classifier. (4) classification results are cached locally. (5) unsafe content is blocked through browser interface intervention. This design enables scalable and near real-time automated moderation while minimizing redundant server computation.

Table 1. Proposed ResNet50–LSTM Model Architecture Parameters

Component	**Layer / Module**	**Input Shape**	**Output Shape**	**Key Parameters / Description**
Input	Video Segment	25 × 224 × 224 × 3	25 frames	1-second clip sampled at 25 fps
Frame Feature Extractor	ResNet-50 Backbone	224 × 224 × 3	7 × 7 × 2048	Pretrained on ImageNet; convolutional layers used for spatial feature extraction
Pooling	Global Average Pooling	7 × 7 × 2048	2048	Converts each frame into a 2048-dimensional feature vector
Sequence Construction	Frame Feature Sequence	25 × 2048	25 × 2048	Combines extracted frame-level features into temporal sequence
Temporal Modeling	LSTM	25 × 2048	128	128 hidden units; captures temporal dependencies across frames
Regularization	Dropout	128	128	Dropout rate = 0.3
Dense Layer	Fully Connected	128	64	ReLU activation
Output Layer	Classification Layer	64	2	Softmax activation; classes: Safe and Unsafe

4 Experiments and Results

4.1 Dataset

Experiments were conducted using a combination of benchmark video datasets and a custom-collected dataset designed for youth-oriented content moderation. The datasets were selected to provide diversity in visual content, including safe, unsafe, live-action, and animated video scenarios relevant to automated video screening.

1. **YouTube-8M Subset**: A curated collection of 5,000 video clips across multiple categories, including violent, adult, and safe content.
2. **NSFW Video Dataset**: 2,800 short clips tagged as safe/unsafe.
3. **Custom Youth Dataset**: 1,500 video samples (cartoons, educational videos, and flagged inappropriate clips in Thai/English) collected for this study.

For preprocessing, all videos were standardized to a spatial resolution of 224 × 224 pixels and resampled at 25 frames per second (fps). Each video was segmented into non-overlapping 1-second clips, resulting in sequences of 25 consecutive frames per sample, which served as input to the ResNet50–LSTM classification framework. For each segment, both video frames and audio transcripts were analyzed. To ensure representativeness, the dataset was split into 80% training, 10% validation, and 10% testing sets. Stratified sampling preserved the ratio of safe/unsafe samples across splits. All models were trained using early stopping and batch normalization to prevent overfitting. The ResNet50–LSTM classification framework were trained end-to-end using the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and 50 epochs. The categorical cross-entropy loss

function was used to minimize classification error. Early stopping was applied based on validation loss to avoid overfitting. All models were implemented using Tensor Flow 2.15.

4.2 Evaluation Metrics

The confusion matrix is a potent technique for assessing the performance of classification results. It provides a detailed representation of predicted and actual classes, aiding in the visualization of classification errors. The matrix is structured such that each row corresponds to the predicted class, and each column represents the original class. The evaluation metrics derived from the confusion matrix include precision, recall, F1 score, and accuracy, calculated using the following equations. Precision is the precision value calculated from (1), Recall is the recall value calculated from (2), F1 score is calculated from (3), and accuracy is calculated from (4).

$$\boldsymbol{Precision} = \frac{TP}{TP+FP} \quad (1)$$

where TP is the true positive value and FP is the false positive value. Precision is the value that measures the number of correct predicting answers by dividing by the total number of images in the dataset.

$$\boldsymbol{Recall} = \frac{TP}{TP+FN} \quad (2)$$

where TP is the true positive value and FN is the false negative value. Recall is the value that measures the number of correct predicting answers for each class and is divided by the total number of that class which is the ground truth.

$$\boldsymbol{F1-Score} = 2\frac{PrecisionxRecall}{Precision+Recall} \quad (3)$$

$$\boldsymbol{Accuracy} = \frac{TP+TN}{N} \quad (4)$$

Where TP is the true positive value, TN is the true negative value, and N is total number of images.

4.3 Experimental results

Table 2. presents the confusion matrix for the ResNet50 + LSTM hybrid model. The system achieved high precision for the “safe” category, while most misclassifications occurred in borderline cases such as mild cartoon violence. Performance comparison across models are show in Table 3.

Table 2: Confusion Matrix for ResNet50 + LSTM Hybrid Model.

Predicted/Actual	**Safe**	**Unsafe**
Safe	1,280	120
Unsafe	95	1,305

Table 3:Performance comparison across models.

Method	**Precision (%)**	**Recall (%)**	**F1-score (%)**	**Accuracy (%)**
Keyword-based Filtering	62.5	58.2	60.3	65.1
Rule-based Metadata Filtering	68.9	63.5	66.1	70.2
CNN (ResNet50 Frames Only)	84.6	81.9	83.2	85.0
LSTM (Temporal Sequence Only)	79.8	76.4	78.1	80.3
ResNet50 + LSTM Hybrid (Proposed)	**93.2**	**91.6**	**92.4**	**92.3**

4.4 Discussions

The proposed ResNet50–LSTM hybrid framework demonstrated strong classification performance for automated youth-oriented video moderation, achieving high precision and recall in distinguishing safe and unsafe content. The confusion matrix indicates that the majority of classification errors occurred in borderline cases, particularly involving mild cartoon violence or stylized animated scenes.

One important explanation for these errors is the domain adaptation challenge commonly observed in computer vision systems. The pretrained ResNet50 model was originally optimized for natural image recognition using ImageNet, while the primary training datasets used in this study (e.g., YouTube-8M and NSFW) predominantly contain live-action visual content. Cartoon media differs significantly from real-world imagery in terms of abstract shapes, exaggerated color schemes, simplified textures, and symbolic representations of harmful actions. As a result, spatial feature representations learned from real-world data may not fully generalize to animated content, increasing the likelihood of false classifications.

Future improvements should address this limitation through domain adaptation strategies such as incorporating animation-specific training datasets, transfer learning with cartoon-focused fine-tuning, and domain generalization methods to improve robustness across heterogeneous media types.

Another practical consideration concerns deployment latency. Although deep learning inference using ResNet50 and LSTM introduces computational overhead, the proposed browser extension minimizes redundant processing through IndexedDB caching of previously classified videos. Only uncached content requires backend inference.

Latency profiling of the prototype system shows an average total processing time of approximately 381 milliseconds per uncached video request, including metadata retrieval, frame preprocessing, server inference, and browser response actions. This operational profile supports near real-time deployment for practical browser-based content moderation.

However, scalability remains a consideration for high-throughput production deployment. Further optimization strategies such as model compression, TensorRT acceleration, edge inference, or lightweight architectures (e.g., MobileNetV3, EfficientNet-lite) could significantly reduce latency in future implementations.

Finally, while the current framework focuses on visual temporal moderation, future multimodal extensions integrating speech transcription, subtitle analysis, and contextual metadata could further improve classification reliability for ambiguous or culturally sensitive content.

Examples of inappropriate videos was shown in Figure. 2.



Figure 2: Examples of inappropriate videos.

5 Conclusion

The increasing dependence of children and adolescents on online video platforms has created a growing demand for effective and scalable content moderation solutions. Traditional manual review methods are difficult to scale and are often unable to cope with the vast volume of multimedia content continuously uploaded to online platforms. This study presented an AI-based automated video screening system for youth-appropriate content using a hybrid deep learning framework that combines a pretrained ResNet-50 Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network. The proposed architecture enables the extraction of high-level spatial features from video frames while simultaneously capturing temporal relationships across video sequences, allowing for more accurate identification of inappropriate visual content.

Experimental evaluation conducted on benchmark datasets and a custom youth-oriented video dataset demonstrated the effectiveness of the proposed framework. The ResNet50–LSTM model achieved superior performance compared with conventional keyword-based filtering, metadata-based screening, and single-model approaches, obtaining 93.2% precision, 91.6% recall, 92.4% F1-score, and 92.3% accuracy. Analysis of the confusion matrix further confirmed the model's capability to distinguish between safe and unsafe content, with most classification errors occurring in challenging borderline cases such as mild cartoon violence and stylized animated scenes.

The integration of the classification model with a browser extension and IndexedDB-based caching mechanism demonstrates the practical feasibility of near real-time automated content moderation. By reducing redundant server-side inference requests and efficiently managing previously classified content, the proposed system provides a scalable solution suitable for deployment in modern digital media environments.

Overall, this research contributes to the advancement of intelligent content moderation systems and E-Business Information Systems by providing a reliable, scalable, and practically deployable framework for protecting youth audiences from inappropriate visual content. Future work will focus on improving robustness across diverse content domains, particularly animated and culturally specific media, as well as exploring multimodal extensions incorporating audio, speech, and textual information to further enhance moderation accuracy and contextual understanding..

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